

# Arithmetic reconstruction and endpoint regularity of the completed $\zeta$ response

Dylan Suvlu

24 September 2026

**Abstract.** The  $\theta$ -correlation companion reduces the sign of the positive-rate Weil kernel to one question. Does an explicit source  $b_*$  lie in the range of a compact Hankel operator  $H_b$ ? Equivalently, is the unique square-integrable solution of a fixed convolution equation causal? We construct the causal solution  $q_c$  directly from arithmetic. With  $\kappa(t) = e^{-t}$  for  $t > 0$ , the response  $u = q_c + \kappa$  has Laplace transform  $U(r) = 2 \frac{\xi(\frac{1}{2}+r)}{(r+1)(\xi+\xi')(\frac{1}{2}+r)}$ . A weighted Volterra equation produces it on positive times, and at each time  $t$  it is a finite sum over integer delays  $\log n \leq t$  with von Mangoldt convolution coefficients. On every finite interval, finite Möbius inversion reconstructs it from its completed output, and conjugation to the Euler edge leaves one fixed forcing equation. Unconditionally, the edge response  $g = e^{-\frac{t}{2}}u$  satisfies  $\|t^k g\|_2 \leq AB^k(k!)^2$ , so  $e^{a\sqrt{t}}g \in L^2$  for some  $a > 0$ . The source  $b_*$  lies in the range of  $|H_b|^{1-\delta}$  for every  $0 < \delta < 1$ , and of no  $|H_b|^{1+\delta}$  with  $\delta > 0$ . Its regularized inverse energy is at most  $C[1 + \log(\frac{1}{\lambda})]^{\frac{1}{2}}$ , with a Gevrey refinement. These bounds leave the first power undecided. Conditionally, if  $U$  has no genuine pole in a half-plane  $\Re r > a$  with  $0 \leq a < \frac{1}{2}$ , then  $e^{\beta t}g \in L^2$  for an explicit  $\beta > 0$ ; poles cancelled by zeros of  $\xi$  do not count. Finally, the Riemann hypothesis is equivalent to the absence of genuine poles in  $\Re r > 0$ , to  $e^{\beta t}g \in L^2$  for every  $0 < \beta < \frac{1}{2}$ , and to  $q_c \in L^2(0, \infty)$ , which is the range condition  $b_* \in \text{Ran } H_b$  itself.

## 1. Introduction

For a convolution equation, square-integrability and one-sided support are different requirements on an inverse. The equation studied here has a unique ordinary full-line  $L^2$  solution, but that solution is not known to be causal. A direct causal construction imposes positive-time support from the outset and initially controls the solution only in an exponentially weighted space. These uniqueness statements concern different classes. The endpoint asks whether the supported solution also belongs to ordinary  $L^2$ , in which case it must coincide with the full-line solution; (101) gives this identification for the actual Hankel range.

This is the one-source inverse problem associated with the  $\theta$ /Hankel range criterion in [1, §6, Theorem “Range and causality”]. There, causality of the ordinary solution is equivalent to actual membership of a fixed source in the range of a compact Hankel operator. Here we construct the supported inverse directly from arithmetic, determine regularity on both sides of the first-power range threshold, and prove a conditional exponential improvement. The construction and estimates do not assume that the exact range endpoint holds.

Arithmetic supplies a causal construction because multiplication of integers becomes addition of delays after taking logarithms. The logarithmic derivative of  $\zeta$  contributes von Mangoldt coefficients, and by time  $t$  only delays  $\log n \leq t$  can have acted. The resulting expansion is therefore finite in the integer variable at each physical time. On a finite interval, the same delay algebra admits Möbius inversion. This gives a reconstruction of the completed response without invoking an infinite Möbius series below its absolute-convergence line.

To identify the response, write

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s), \quad M(r) = \frac{\xi(\frac{1}{2}+r)}{\xi(\frac{1}{2})}, \quad \ell = \frac{M'}{M}. \quad (1)$$

Let  $p$  be the normalized even  $\theta$  density with moment generating function  $M$ . Put  $a(t) = (1-t)p(t)$ ,  $b(t) = (1+t)p(t)$  and  $\kappa(t) = e^{-t}\mathbf{1}_{t>0}$ . The equation is  $a * q = \kappa * b$ . Its bilateral Laplace symbols are  $M + M'$  and  $M - M'$ , so the supported solution constructed in Section 2 satisfies, on its initial half-plane of convergence,

$$\hat{q}_c(r) = \frac{1 - \ell(r)}{(r+1)(1 + \ell(r))}. \quad (2)$$

Adding the fixed source gives the response

$$u = q_c + \kappa, \quad U(r) = \hat{u}(r) = \frac{2}{(r+1)(1 + \ell(r))} = 2 \frac{\xi(\frac{1}{2}+r)}{(r+1)(\xi + \xi')(\frac{1}{2}+r)}. \quad (3)$$

A denominator zero is a genuine response pole when the numerator does not cancel it. At a zero of  $\xi$ , this quotient has a removable zero instead. The physical construction, rather than meromorphic continuation, justifies the initial transform identity.

The useful coordinate for unconditional decay is the Euler edge. In the  $\zeta$  variable  $s_\zeta = \frac{1}{2} + r$ , that edge is  $\Re s_\zeta = 1$ , or  $\Re r = \frac{1}{2}$ . Define  $g = e^{-\frac{t}{2}}u$ . Then

$$q_c = e^{\frac{t}{2}}g - \kappa. \quad (4)$$

Since  $\kappa \in L^2$ , the ordinary causal-preimage endpoint is exactly  $e^{\frac{t}{2}}g \in L^2$ . A square-root-exponential weight for  $g$ , or a positive linear exponential weight smaller than  $\frac{1}{2}$ , gives useful decay without reaching that endpoint. We organize the results into three stages.

**Construction and finite inversion.** We split the response denominator into an invertible archimedean baseline and arithmetic feedback. A weighted Volterra contraction gives the supported inverse; regrouping its convergent series produces the finite-time von Mangoldt expansion. The completed output is inverted by finite Möbius shifts after differential/Volterra preprocessing. Conjugation to the Euler edge removes the weight parameter from this reconstruction and leaves one fixed forcing equation. Its centering preserves the cancellation between the constant and transient Möbius terms. The generic inverse cost and the special source response are kept distinct.

**Unconditional regularity.** The edge response satisfies a Gevrey-2 moment bound,

$$\|t^k g\|_2 \leq AB^k(k!)^2 \quad (k \geq 0), \quad (5)$$

and hence  $e^{a_0\sqrt{t}}g \in L^2$  for some  $a_0 > 0$ . The same estimate holds for the related edge outputs specified in Section 6. Separately, the distinguished Hankel source  $b_*$  belongs to  $\text{Ran}|H_b|^{1-\delta}$  for every  $0 < \delta < 1$ , and to none of the ranges  $\text{Ran}|H_b|^{1+\delta}$  for  $\delta > 0$ . These are different regularity statements. The moment hierarchy controls long-time decay. The fractional range result combines approximate preimages with the logarithmic initial behavior of the causal inverse; the exact first-power range remains open. For the original regularized inverse energy of [1, Theorem “Inverse energy and rank-one detection”],

$$\mathcal{E}(\lambda) = \left\langle b_*, \left( (H_b^2 + \lambda I)^{-1} b_* \right) \right\rangle, \quad (6)$$

we prove  $\mathcal{E}(\lambda) \leq C[1 + \log(\frac{1}{\lambda})]^{\frac{1}{2}}$  in (105). The Gevrey weight sharpens this bound in (116). Both bounds allow divergence as  $\lambda$  tends to zero and therefore leave the first-power range endpoint undecided.

**Conditional exponential improvement.** If the response has no genuine poles in a fixed half-plane  $\Re r > a$ ,  $0 \leq a < \frac{1}{2}$ , we prove  $e^{\beta_a t} g \in L^2$  for an explicit  $\beta_a > 0$ . The Hardy conclusion holds in the smaller half-plane  $\Re r > \frac{1}{2} - \beta_a$ , with the quantitative loss given in Section 8. Factoring common zeros, Harnack propagation and three-lines interpolation provide the reciprocal growth control needed for this conclusion. The pole gap is a hypothesis, not a consequence of the regularity theorem.

The derivative pair  $\xi, \xi'$  and its half-plane criteria occur in Lagarias [2, 3] and Suzuki [4]. The completion multiplier is that of [5]. For the unconditional estimates we use the prime number theorem bound of Johnston and Yang [6] and the zero-free region of Mossinghoff and Trudgian [7]. These arithmetic inputs give stretched-exponential control at the edge; the conditional theorem addresses the additional growth estimate needed to return into a fixed shifted Hardy half-plane.

Section 2 constructs the causal solution. Sections 3–5 treat finite inversion, edge forcing and centering. Sections 6–7 prove the moment and fractional-range results. Section 8 proves the pole-gap theorem. Section 9 distinguishes its existential conclusion from the all-weight endpoint and shows that the range endpoint itself is equivalent to the Riemann hypothesis.

## 2. Constructing the causal arithmetic response

We use ordinary physical  $L^2$  and conjugate-linear-first inner products. The convolutions defining the supported reconstruction below are causal. The Laplace transform is  $\hat{f}(r) = \int_0^\infty e^{-rt} f(t) dt$ . Let

$$M(r) = \frac{\xi(\frac{1}{2} + r)}{\xi(\frac{1}{2})}, \quad \ell(r) = \frac{M'(r)}{M(r)}, \quad \kappa(t) = e^{-t}. \quad (7)$$

We first work in an absolutely convergent Euler half-plane. There

$$\ell(r) = \frac{1}{r + \frac{1}{2}} + \frac{1}{r - \frac{1}{2}} - \frac{\log(\pi)}{2} + \frac{\psi\left(\frac{r+\frac{1}{2}}{2}\right)}{2} - \sum_{n \geq 2} \Lambda(n) n^{-r-\frac{1}{2}}. \quad (8)$$

The construction separates inversion from arithmetic feedback. We split  $1 + \ell = B_0 - \hat{\nu}$  into an explicitly invertible logarithmic baseline and a signed prime/ $\Gamma$  term. A positive kernel  $\rho$  inverts the baseline. At one sufficiently large exponential weight,  $\rho * \nu$  is a contraction, so its causal Neumann series converges. Only after this convergence is established do we regroup the prime atoms by their integer delays.

Put  $b_0 = 1 - \frac{\log(2\pi)}{2} > 0$  and  $B_0(r) = b_0 + \frac{\log(1+r)}{2}$ , using the principal logarithm. Define the archimedean response

$$d_\Gamma(r) = 1 + \frac{1}{r + \frac{1}{2}} + \frac{1}{r - \frac{1}{2}} - \frac{\log(\pi)}{2} + \frac{\psi\left(\frac{r+\frac{1}{2}}{2}\right)}{2}. \quad (9)$$

The digamma and logarithm integrals, combined before integration at the origin, give

$$c_\Gamma(t) = 2 \cosh\left(\frac{t}{2}\right) + \frac{e^{-t}}{2t} - \frac{e^{-\frac{t}{2}}}{1 - e^{-2t}}, \quad c_\Gamma(0) = \frac{5}{4}, \quad (10)$$

$$\hat{c}_\Gamma(r) = d_\Gamma(r) - B_0(r). \quad (11)$$

Indeed the last two summands in (10) have transform  $\frac{\left[\psi\left(\frac{r+\frac{1}{2}}{2}\right) - \log\left(\frac{r+1}{2}\right)\right]}{2}$ . The singularities cancel. Direct expansion gives  $c_\Gamma(t) = \frac{5}{4} + 13 \frac{t}{48} + O(t^2)$ . The elementary inequalities

$$e^{\frac{t}{2}} - \frac{1}{4} \leq c_\Gamma(t) < 2 \cosh\left(\frac{t}{2}\right) \quad (12)$$

follow from  $(1 - e^{-2t})^{-1} \leq 1 + (2t)^{-1}$  and  $e^{-\frac{t}{2}} - e^{-t} \leq \frac{t}{2}$ .

Let  $V = \sum_{n \geq 2} \Lambda(n) n^{-\frac{1}{2}} \delta_{\log n}$  and  $\nu = V - c_\Gamma(t) dt$ . Then

$$1 + \ell(r) = B_0(r) - \hat{\nu}(r). \quad (13)$$

A positive inverse of the scalar baseline is

$$\rho(t) = 2e^{-t} \int_0^\infty e^{-2b_0 a} \frac{t^{a-1}}{\Gamma(a)} da. \quad (14)$$

Tonelli and the  $\Gamma$  integral give  $\hat{\rho}(r) = \frac{1}{B_0(r)}$  for real  $r \geq 0$  and  $\|\rho\|_1 = \frac{1}{b_0}$ .

Use the weighted convolution algebra with norm  $\|f\|_{1,4} = \int e^{-4t} |f(t)| dt$ . The feedback  $k = \rho * \nu$  obeys

$$\|k\|_{1,4} \leq \frac{\sum_{n \geq 2} \Lambda(n) n^{-\frac{9}{2}} + \hat{c}_\Gamma(4)}{B_0(4)} < \frac{7}{10}. \quad (15)$$

For explicit bounds,  $B_0(4) > \frac{61}{70}$ ,  $\hat{c}_\Gamma(4) \leq \frac{32}{63}$ , and

$$\sum_{n \geq 2} \Lambda(n) n^{-\frac{9}{2}} \leq \sum_{n \geq 2} \log(n) n^{-4} \leq 5 \frac{\log(2)}{48} + \frac{1}{72}. \quad (16)$$

With  $\log 2 < \frac{7}{10}$  their ratio is less than  $\frac{5995}{8784}$ . No estimate in an unproved zero-free half-plane has been used.

**Theorem 1** (Supported reconstruction).

The convergent weighted convolution series

$$r_c = \sum_{j \geq 0} k^{*j} * \rho, \quad q_c = -\kappa + 2\kappa * r_c, \quad u = q_c + \kappa \quad (17)$$

defines a causal locally square-integrable response with  $\|e^{-4t} q_c\|_2 < 3$ . For  $\Re r \geq 4$ ,

$$U(r) := \hat{u}(r) = \frac{2}{(r+1)(1+\ell(r))} = 2 \frac{\xi(\frac{1}{2} + r)}{(r+1)(\xi + \xi')(\frac{1}{2} + r)}. \quad (18)$$

There are explicitly defined archimedean kernels  $U_k$  for which

$$q_c(t) = -e^{-t} + \sum_{n \leq e^t} n^{-\frac{1}{2}} \sum_{k \geq 0} \Lambda^{*D^k}(n) U_k(t - \log n). \quad (19)$$

Both sums in (19) are finite at every fixed time.

**Proof.** Equation (15) proves absolute convergence of (17) in the weighted  $L^1$  algebra. Weighted Young's inequality gives

$$\|e^{-4t} q_c\|_2 \leq \frac{1}{\sqrt{10}} \left( 1 + \frac{2}{B_0(4)(1 - \|k\|_{1,4})} \right) < 3. \quad (20)$$

Transforming that convergent series and using (13) proves (18). It also gives

$$\hat{q}_c(r) = \frac{1 - \ell(r)}{(r+1)(1 + \ell(r))}. \quad (21)$$

For the delay formula separate the smooth feedback:

$$r_\Gamma = \sum_{j \geq 0} (-\rho * c_\Gamma)^{*j} * \rho, \quad U_k = 2\kappa * r_\Gamma^{*(k+1)}. \quad (22)$$

The same positive majorants show convergence and  $\|r_\Gamma * V\|_{1,4} < 1$ . Thus the absolutely convergent series may be regrouped as  $r_c = \sum_{k \geq 0} r_\Gamma^{*(k+1)} * V^{*k}$ . Here

$$V^{*k} = \sum_{n \geq 1} \Lambda^{*D^k}(n) n^{-\frac{1}{2}} \delta_{\log n}. \quad (23)$$

The zeroth Dirichlet convolution is supported at 1. For  $n > 1$ , a nonzero  $k$ -fold coefficient requires  $k \leq \Omega(n)$ , where  $\Omega(n)$  counts prime factors with multiplicity. Causality requires  $n \leq e^t$ . This proves finiteness and (19). Each  $U_k(0) = 0$ , so a term born at equality contributes zero.  $\square$

This supported solution is related to the  $\theta$  convolution equation. Write  $\xi(s) = s(s-1)\pi^{-\frac{s}{2}}\Gamma(\frac{s}{2})\frac{\zeta(s)}{2}$  and  $f_0 = \xi(\frac{1}{2})$ . The normalized  $\theta$  density is, for  $t \geq 0$ ,

$$p(t) = \frac{1}{f_0} \sum_{n \geq 1} (4\pi^2 n^4 e^{9\frac{t}{2}} - 6\pi n^2 e^{5\frac{t}{2}}) e^{-\pi n^2 e^{2t}}, \quad (24)$$

with its smooth even continuation. Poisson summation and the Mellin identity give its moment generating function  $M$ . Set  $a(t) = (1-t)p(t)$ ,  $b(t) = (1+t)p(t)$  on the full line. Their bilateral Laplace transforms, with kernel  $e^{-rt}$ , are  $M + M'$  and  $M - M'$ . Equation (21) gives

$$a * q_c = \kappa * b. \quad (25)$$

The calculation is justified in weighted full-line  $L^2$  on a common line to the right of 4; the  $\theta$  kernels have all exponential moments. Fourier uniqueness makes  $q_c$  the unique supported solution in that class. Ordinary full-line  $L^2$  uniqueness, when applicable, is a separate statement.

## 2.1. Möbius transport of the Hankel equation

The same supported solution can be expressed with a single archimedean seed and its von Mangoldt shifts. This change of kernels preserves ordinary Hankel preimages. It is separate from the finite-time reconstruction of the completed output in Section 3.

Define on the full line

$$\varphi_\theta(u) = (4\pi^2 u^4 - 6\pi u^2) e^{-\pi u^2}, \quad g_0(t) = \frac{e^{\frac{t}{2}} \varphi_\theta(e^t)}{f_0}, \quad (26)$$

$$j(t) = \sum_{n \geq 2} \Lambda(n) n^{-\frac{1}{2}} g_0(t + \log n), \quad (27)$$

and put

$$\tilde{a}(t) = (1-t)g_0(t) + j(t), \quad \tilde{b}(t) = (1+t)g_0(t) - j(t). \quad (28)$$

For a kernel  $k$ , write

$$(H_k f)(t) = \int_0^\infty k(t+u) f(u) du, \quad t > 0. \quad (29)$$

The positive tails of all these kernels make the displayed Hankel operators Hilbert–Schmidt.

**Proposition (The same Hankel preimage).** For every  $q \in L^2(0, \infty)$ ,

$$H_b q = H_a \kappa \iff H_{\tilde{b}} q = H_{\tilde{a}} \kappa. \quad (30)$$

**Proof.** The  $\theta$  expansion is

$$p(t) = \sum_{n \geq 1} n^{-\frac{1}{2}} g_0(t + \log n). \quad (31)$$

Apply the kernel sum

$$\mathcal{Q}k(t) = \sum_{d \geq 1} \mu(d) d^{-\frac{1}{2}} k(t + \log d). \quad (32)$$

The divisor identities

$$\sum_{d|n} \mu(d) = \mathbf{1}_{n=1}, \quad \sum_{d|n} \mu(d) \log d = -\Lambda(n) \quad (33)$$

give

$$\mathcal{Q}p = g_0, \quad \mathcal{Q}(tp) = tg_0 - j, \quad \mathcal{Q}a = \tilde{a}, \quad \mathcal{Q}b = \tilde{b}. \quad (34)$$

The inverse Euler sums give

$$a(t) = \sum_{n \geq 1} n^{-\frac{1}{2}} \tilde{a}(t + \log n), \quad b(t) = \sum_{n \geq 1} n^{-\frac{1}{2}} \tilde{b}(t + \log n). \quad (35)$$

These rearrangements are justified on kernels, including in the Hankel Hilbert–Schmidt norm

$$\|k\|_{\text{H,HS}}^2 = \int_0^\infty t |k(t)|^2 dt. \quad (36)$$

Indeed a shift by  $\log n$  of any positive-tail kernel above is bounded in this norm by a polynomial in  $n, \log n$  times  $e^{-cn^2}$ . The seed formula gives this directly; the  $\theta$  synthesis and the sum defining  $j$  give the same estimate by absolute summation. It also holds locally uniformly in  $t$ . Thus both the forward Möbius series and the inverse Euler series converge absolutely in this norm, as do the double sums used in the divisor identities.

For a fixed ordinary  $L^2$  vector, Hilbert–Schmidt convergence permits passing these sums through the Hankel integral. Applying the forward sums to the output variable gives the forward implication in (30). Applying the inverse sums to the transported equation gives the converse. This proves the assertion for the same vector  $q$ , without an infinite arithmetic operator acting boundedly on all of  $L^2$ .  $\square$

The transform identification keeps track of reflection. For kernels use the plus-sign bilateral transform

$$\mathcal{B}_+ k(r) = \int_{\mathbb{R}} e^{rt} k(t) dt, \quad (37)$$

and for a supported response retain the minus-sign Laplace transform. Put

$$\xi_\infty(z) = \frac{1}{2} z(z-1) \Gamma_{\mathbb{R}}(z), \quad F(r) = \frac{\xi_\infty(\frac{1}{2} + r)}{f_0}, \quad V(r) = \sum_{n \geq 2} \Lambda(n) n^{-\frac{1}{2}-r}. \quad (38)$$

Here  $\Gamma_{\mathbb{R}}(z) = \pi^{-\frac{z}{2}}\Gamma(\frac{z}{2})$ . Mellin integration and the shift sums give, on  $\Re r > \frac{1}{2}$ ,

$$\mathcal{B}_+ g_0 = F, \quad \mathcal{B}_+ j = FV, \quad \ell = \frac{F'}{F} - V. \quad (39)$$

It follows that

$$\mathcal{B}_+ \tilde{a} = F(1 - \ell), \quad \mathcal{B}_+ \tilde{b} = F(1 + \ell). \quad (40)$$

For  $\check{k}(t) = k(-t)$ , the reflected equation is

$$\check{\tilde{b}} * q_c = \check{\tilde{a}} * \kappa. \quad (41)$$

On a common safe line  $\Re r > 4$ , its transform is exactly

$$\hat{q}_c(r) = \frac{1 - \ell(r)}{(r + 1)(1 + \ell(r))}, \quad (42)$$

so weighted Fourier uniqueness identifies the same supported response constructed above. Absolute kernel integrability with this weight justifies the transformed convolution. In general  $\check{\tilde{b}}$  is not  $\tilde{a}$ .

The seed  $g_0$  is distinct from the edge response  $g = e^{-\frac{t}{2}}(q_c + \kappa)$  used below, and (28) are not the forcing kernels of Section 4. The same-preimage identity preserves the actual range question and the norm of its solution. Regularized inverse energies at positive parameter, and fractional powers of the two Hankel operators, remain different objects.

## 2.2. The initial logarithmic boundary

Put  $L_t = \log(\frac{1}{t}) + 2b_0$ . Since  $\frac{1}{\Gamma(a)} = a + \gamma a^2 + O(a^3)$  as  $a$  decreases to zero, splitting the shape integral in (14) at a fixed small value gives

$$\rho(t) = \frac{2}{tL_t^2}(1 + O(L_t^{-1})), \quad \int_0^t \rho(v) dv = \frac{2}{L_t} + O(L_t^{-2}). \quad (43)$$

The large-shape remainder is smaller by  $\Gamma$  growth. On a small interval  $c_{\Gamma}$  is bounded, and its convolution perturbation gives  $\|r_{\Gamma} - \rho\|_{L^1(0,t)} = O(tL_t^{-2})$  by its local geometric series. Before the first prime delay only  $U_0$  is active. Therefore

$$q_c(t) = -1 + \frac{4}{L_t} + O(L_t^{-2}) \quad (t \downarrow 0). \quad (44)$$

It has a finite right limit but no positive Hölder exponent there. This local fact will distinguish the first Hankel power from every larger power.

## 3. Finite Möbius reconstruction

On  $L^2(0, R)$  define the killed causal shift  $T_a f(t) = f(t - a)$  for  $t \geq a$  and zero otherwise. Then  $T_a T_b = T_{a+b}$  and  $T_a = 0$  for  $a \geq R$ . Fix  $\sigma > 0$ , and put  $\alpha_{\sigma} = \frac{1}{2} + \sigma$ ,  $\beta = \frac{1}{2} - \sigma$ . Define

$$P_{\sigma,R} = \sum_{n \leq e^R} n^{-\alpha_{\sigma}} T_{\log n}, \quad Q_{\sigma,R} = \sum_{n \leq e^R} \mu(n) n^{-\alpha_{\sigma}} T_{\log n}. \quad (45)$$

All sums are finite; the possible last term  $T_R$  is zero as an  $L^2$  operator. Divisor collection and unique factorization give

$$P_{\sigma,R} Q_{\sigma,R} = Q_{\sigma,R} P_{\sigma,R} = I, \quad (46)$$

$$Q_{\sigma,R} = \prod_{p \leq e^R} (I - p^{-\alpha_\sigma} T_{\log p}). \quad (47)$$

The inverse product for  $P$  consists of terminating geometric sums. These are identities in the killed shift algebra, independent of infinite Dirichlet convergence.

The critical completed multiplier is

$$m(r) = 2 \frac{\xi(\frac{1}{2} + r)}{(r + \frac{5}{2})(r + \frac{3}{2})\Gamma_{\mathbb{R}}(r + \frac{5}{2})} = 2\pi \left(r - \frac{1}{2}\right) \frac{\zeta(r + \frac{1}{2})}{(r + \frac{3}{2})(r + \frac{5}{2})}, \quad (48)$$

where  $\Gamma_{\mathbb{R}}(s) = \pi^{-\frac{s}{2}}\Gamma(\frac{s}{2})$ .  $\Gamma$  recurrence proves the second equality. The point  $r = \frac{1}{2}$  is removable, with value  $\frac{\pi}{3}$ . The causal kernel of completion is the locally finite sum

$$K(t) = 2\pi \sum_{n \leq e^t} n^{-\frac{1}{2}} \left[ -2e^{-3\frac{t - \log n}{2}} + 3e^{-5\frac{t - \log n}{2}} \right]. \quad (49)$$

For  $J = \lfloor e^t \rfloor$ , summing the two powers gives

$$K(t) = 2\pi e^{-5\frac{t}{2}} J(J+1) \left( J + \frac{1}{2} - e^t \right), \quad (50)$$

and hence  $|K(t)| \leq 2\pi e^{-\frac{t}{2}}$ .

This is the forward causal realization of the kernel in the operator companion [5, Theorem “Completed half-line operator”]. That paper uses backward shifts; the killed causal shifts here are their half-line adjoints. At the critical parameter the kernel is real, so the corresponding half-line convolution operators are adjoints. On a finite interval, reflection interchanges the terminal absorbing boundary there with the initial boundary used here.

Let  $\eta = K * u$  be the completed causal output (distinct from the ordinary full-line solution in Paper II),  $f_\sigma = e^{-\sigma t}u$  and  $\eta_\sigma = e^{-\sigma t}\eta$ . Introduce

$$V_\sigma f = \left[ -2e^{-(\sigma + \frac{3}{2})t} + 3e^{-(\sigma + \frac{5}{2})t} \right] * f, \quad (51)$$

$$\mathcal{B}_\sigma \eta = \eta' + \left( \sigma + \frac{9}{2} \right) \eta + 6(e^{\beta t} * \eta). \quad (52)$$

On each finite interval,  $V_\sigma$  maps  $L^2$  bijectively onto  $H_{\text{in}}^1 = \{\eta \in H^1(0, R) : \eta(0) = 0\}$ , with inverse  $\mathcal{B}_\sigma$ . To verify this, its kernel has value 1 at zero, so differentiation gives  $(V_\sigma f)' = f + V_\sigma' * f$ ; the rational identity

$$\left( r + \frac{3}{2} \right) \left( r + \frac{5}{2} \right) = \left( r - \frac{1}{2} \right) \left( r + \frac{9}{2} \right) + 6 \quad (53)$$

proves the two compositions first on smooth initial-zero functions and then by boundedness on these finite-window spaces. Equivalently they follow from the two first-order Volterra equations for the exponential kernels.

**Theorem 2** (Assembled finite inverse).

The finite completed map is  $2\pi V_\sigma P_{\sigma,R} : L^2(0, R) \rightarrow H_{\text{in}}^1$ , and

$$2\pi f_{\sigma,R} = Q_{\sigma,R} [\mathcal{B}_\sigma \eta_\sigma]|_{(0,R)}. \quad (54)$$

The derivative is intrinsic to the interval, with no terminal condition.

**Proof.** Summing the kernel (49) on a finite interval gives the displayed factorization directly. Finite causal convolutions commute. Apply the inverse of  $V_\sigma$  and the divisor identity (47). An

unbounded derivative is commuted only on its initial-zero domain. Restriction of the actual causal output before  $R$  equals completion of the input restricted before  $R$ , by causality. A sharp cutoff derivative would instead introduce a terminal  $\delta$  and is not used.  $\square$

**Proposition 3** (Generic conditioning of the Möbius map).

For fixed  $\sigma > 0$ , as  $R$  tends to infinity,

$$\|Q_{\sigma,R}\| \underset{\sigma}{\asymp} \begin{cases} e^{(\frac{1}{2}-\sigma)R} & 0 < \sigma < \frac{1}{2} \\ 1 + R & \sigma = \frac{1}{2} \\ 1 & \sigma > \frac{1}{2} \end{cases}. \quad (55)$$

**Proof.** Let  $Q_{\text{abs}} = \sum \mu(n)^2 n^{-\alpha_\sigma} T_{\log n}$ . Pointwise domination gives  $\|Q\| \leq \|Q_{\text{abs}}\|$ . Conversely modulation by  $e^{i\omega t}$  conjugates each shift by  $e^{-i\omega \log n}$ . The finite set of prime logarithms is rationally independent, by unique factorization. Its continuous torus orbit approaches the phase  $\pi$  at every prime simultaneously: average a smooth nonnegative bump around that torus point; its absolutely convergent Fourier series has all nonconstant time averages zero and positive constant average. For squarefree  $n$  the resulting phases make  $\mu(n)e^{-i\omega \log n} \rightarrow 1$ . Finitely many coefficients imply operator-norm convergence of these conjugates to  $Q_{\text{abs}}$ . Thus their norms are equal.

The upper bound follows from  $\sum_{n \leq e^R} n^{-\alpha_\sigma} \leq 1 + \int_0^R e^{(1-\alpha_\sigma)t} dt$ . For a matching lower bound, the number of squarefree integers up to  $x$  is at least  $\frac{x}{4} - 1$ : the union bound on square divisors uses  $\sum_{d \geq 2} d^{-2} \leq \frac{3}{4}$ . Thus  $(\frac{x}{8}, x]$  contains at least  $\frac{x}{16}$  squarefree integers for  $x \geq 16$ . Applied to  $h = \mathbf{1}_{(0, \log 8)}$ , this gives  $(Q_{\text{abs}}h)(t) \geq \frac{e^{(1-\alpha_\sigma)t}}{16}$  for  $t \geq \log 16$ . Integration on  $(R-1, R)$  proves the subcritical lower order.

At  $\alpha_\sigma = 1$ , each disjoint multiplicative block contributes a fixed positive amount to  $\sum \frac{\mu(n)^2}{n}$ . Testing  $h = \mathbf{1}_{(0,R)}$  gives output squared norm bounded below by a constant times  $R^3$  and input squared norm  $R$ . Above that line, a unit input in a final interval of length less than  $\log 2$  has  $Qh = h$ , giving norm at least 1.  $\square$

The maximizing modulations may depend on  $R$ . This proposition describes the inverse on arbitrary data, while (54) applied to the one fixed arithmetic trajectory preserves its signed assembly.

## 4. A single Euler-edge forcing equation

The finite inverse still depends on the physical weight  $\sigma$ . Conjugating response and output to the Euler boundary absorbs that dependence into one fixed source equation. The spectral coordinates satisfy

$$s_\zeta = \frac{1}{2} + r = 1 + z, \quad z = r - \frac{1}{2}. \quad (56)$$

The Euler edge  $\Re s_\zeta = 1$  is  $\Re r = \frac{1}{2}$ , or  $\Re z = 0$ . The functional-equation critical line is instead  $\Re s_\zeta = \frac{1}{2}$ , or  $\Re r = 0$ . In this section “edge” refers to the Euler edge.

| Object                     | Role                                               |
|----------------------------|----------------------------------------------------|
| $q_c$                      | Supported solution constructed in a weighted space |
| $u = q_c + \kappa$         | Response with Laplace transform $U$                |
| $g = e^{-\frac{t}{2}}u$    | Response conjugated to the Euler edge              |
| $\eta = K * u$             | Completed causal output                            |
| $v = e^{-\frac{t}{2}}\eta$ | Edge-conjugated output                             |

|                                 |                                                   |
|---------------------------------|---------------------------------------------------|
| $\mathcal{H}$                   | Euler-edge forcing before finite Möbius inversion |
| $\mathcal{R} = \mathcal{H} - c$ | Centered transient, defined in Section 5          |

Set

$$g(t) = e^{-\frac{t}{2}}u(t), \quad v(t) = e^{-\frac{t}{2}}\eta(t), \quad (57)$$

$$\mathcal{H}(t) = v'(t) + 5v(t) + 6 \int_0^t v(s) \, ds. \quad (58)$$

For finite windows, define  $P_{e,R} = \sum_{n \leq e^R} n^{-1} T_{\log n}$  and  $Q_{e,R} = \sum_{n \leq e^R} \mu(n) n^{-1} T_{\log n}$ . Directly,

$$T_{\log n}(e^{\beta t} F(t)) = e^{\beta t} n^{-\beta} T_{\log n} F(t). \quad (59)$$

Since  $\frac{1}{2} + \sigma + \beta = 1$ , (54) becomes

$$\mathcal{H}_R = 2\pi P_{e,R} g_R, \quad 2\pi g_R = Q_{e,R} \mathcal{H}_R. \quad (60)$$

The coefficient 5 in (58) is  $\beta + \sigma + \frac{9}{2}$ . Thus this bracket is independent of  $\sigma$ . The Euler-edge forcing  $\mathcal{H}$  is the one function on which every subcritical inverse acts after conjugation; studying its equation now avoids carrying a different forcing for each weight.

#### 4.1. The physical transform on the Euler edge

Write

$$L_0(z) = 1 + \left( \frac{\xi'}{\xi} \right) (1+z), \quad G(z) = \frac{2}{(z + \frac{3}{2}) L_0(z)}. \quad (61)$$

The ratio is always interpreted after removable factors are cancelled. It initially agrees with the Laplace transform of  $g$  on  $\Re z \geq \frac{7}{2}$ , by (18).

The even order-one function  $M$  has its paired genus-zero product in  $r^2$ . Its logarithmic derivative is the locally convergent sum of paired reciprocals over its zeros. Euler nonvanishing and the functional equation place those zeros in  $|\Re r| \leq \frac{1}{2}$ . Hence

$$\Re(1 + \ell(r)) \geq 1 \quad \left( \Re r > \frac{1}{2} \right). \quad (62)$$

Each reciprocal has positive real part there; the real-part sum is absolutely convergent. It follows that  $|G(z)| \leq \frac{2}{|z + \frac{3}{2}|}$  for  $\Re z > 0$ . The Laplace–Hardy theorem [8, Theorems 1.1–1.3, p. 972] supplies a causal  $L^2$  inverse; uniqueness on a common safe line identifies it with the constructed  $g$ .

We will need ordinary edge moments. Let  $\Psi(x) = \sum_{n \leq x} \Lambda(n)$  and  $E_0(t) = e^{-t} \Psi(e^t) - 1$ , so  $E_0(0) = -1$ . The unconditional prime number theorem estimate [6, Theorem 1.1, (1.3)]

$$|E_0(t)| \leq 9.39 t^{1.515} e^{-0.8274\sqrt{t}}, \quad t \geq \log 2, \quad (63)$$

gives all polynomial  $L^1$  and  $L^2$  moments. Stieltjes integration by parts in  $\Re z > 0$ , including the initial trace, yields

$$\sum_{n \geq 2} \Lambda(n) n^{-1-z} - \frac{1}{z} = 1 + (z+1) \hat{E}_0(z). \quad (64)$$

The digamma recurrence therefore gives

$$L_0(z) = d_0(z) - 1 - (z+1)\hat{E}_0(z), \quad (65)$$

$$d_0(z) = 1 - \frac{\log(\pi)}{2} + \frac{\psi(\frac{z+3}{2})}{2}. \quad (66)$$

Differentiate (61) once. With  $a = \frac{3}{2}$ ,

$$G' = -\frac{2}{(z+a)^2 L_0} - 2 \frac{[d'_0 - \hat{E}_0 - (z+1)\hat{E}'_0]}{(z+a)L_0^2}. \quad (67)$$

Every term is in the right-half-plane  $H^2$ :  $|\frac{1}{L_0}| \leq 1$ ,  $d'_0$  is bounded by its trigamma series,  $\hat{E}_0, \hat{E}'_0$  have the  $L^2$  bounds supplied by (63), and  $|\frac{z+1}{z+a}| \leq 1$ . The remaining rational factors have uniform vertical  $L^2$  norms. Fourier differentiation and safe-line uniqueness prove  $tg \in L^2$ , hence  $g \in L^1$ .

The weighted kernel  $K_0(t) = e^{-\frac{t}{2}}K(t)$  obeys  $|K_0(t)| \leq 2\pi e^{-t}$ . Thus  $v = K_0 * g \in L^1$ . Its transform is

$$V(z) = m_0(z)G(z), \quad m_0(z) = m\left(\frac{1}{2} + z\right) = 2\pi z \frac{\zeta(1+z)}{(z+2)(z+3)}. \quad (68)$$

The bounds  $|zG(z)| \leq 2$  and  $|G(z)| \leq \frac{4}{3}$  show that  $V, zV$  are  $H^2$ , since  $m_0$  is the transform of an  $L^2$  causal kernel. Therefore  $v \in H^1$ , with initial trace zero. The two derivations of the bracket transform now agree:

$$\hat{\mathcal{H}}(z) = \left(z + 5 + \frac{6}{z}\right)V(z) = 2\pi\zeta(1+z)G(z), \quad \Re z > 0. \quad (69)$$

The first follows from (58) and  $v(0) = 0$ ; the second also follows by absolutely summing  $P_e g$  in this half-plane. No infinite subcritical Möbius inverse is involved.

## 4.2. An independently specified forcing

Let  $\rho_\Gamma(t) = \frac{e^{-3t}}{1-e^{-2t}}$  and set

$$a_\Gamma(t) = e^{-t} \left[ d_0(0) - 1 + \int_0^t (1 - e^s) \rho_\Gamma(s) ds + \int_t^\infty \rho_\Gamma(s) ds \right], \quad \mathcal{A} = a_\Gamma - E_0. \quad (70)$$

Let  $w(t) = 2(e^{-t} - e^{-3\frac{t}{2}})$  and define the fixed arithmetic forcing

$$W(t) = 4\pi(P_e w)(t) = 8\pi \left[ J e^{-t} - e^{-3\frac{t}{2}} \sum_{n=1}^J \sqrt{n} \right], \quad J = \lfloor e^t \rfloor. \quad (71)$$

**Proposition 4** (Fixed edge forcing).

For  $\mathcal{A}$  in (70), one has  $\mathcal{A} \in L^1 \cap L^2$  and

$$\hat{\mathcal{A}}(z) = \frac{L_0(z)}{z+1}, \quad \widehat{W}(z) = 4\pi \frac{\zeta(1+z)}{(z+1)(z+\frac{3}{2})}, \quad \Re z > 0. \quad (72)$$

The constructed edge functions satisfy

$$\mathcal{A} * g = 2w, \quad \mathcal{A} * \mathcal{H} = W. \quad (73)$$

The second equation determines  $\mathcal{H}$  uniquely among causal locally integrable functions  $F$  for which  $e^{-ct}F(t) \in L^1(0, \infty)$  for some finite real  $c$ . Moreover  $W$  is continuous and  $W(t) = 8\frac{\pi}{3} + O(e^{-t})$ .

**Proof.** The logarithmic singularity of  $a_\Gamma$  at zero is integrable and square-integrable, while its tail decays exponentially. Thus  $\mathcal{A} \in L^1 \cap L^2$  by (63). The digamma identity

$$d_0(z) - d_0(0) = \int_0^\infty \rho_\Gamma(s)(1 - e^{-zs}) ds \quad (74)$$

and absolute Fubini after combining the difference prove the stated transform of  $\mathcal{A}$ . The transform of  $w$  is  $\frac{1}{[(z+1)(z+\frac{3}{2})]}$ . Multiplication by (61) gives  $\mathcal{A} * g = 2w$ , and (69) gives the equation for  $\mathcal{H}$ . These are ordinary convergent transforms in  $\Re z > 0$ , so Laplace uniqueness proves the physical identities. The finite sum for  $W$  follows directly from  $P_e w$ ;  $w(0) = 0$  makes it continuous when a new integer term enters. Integral comparison of the square-root sum gives its asymptotic constant.

For uniqueness, take the transform of the difference of two causal solutions on a common far-right line. There  $\hat{\mathcal{A}}$  is nonzero by (62). The difference transform is therefore zero, and Laplace uniqueness applies.  $\square$

The forcing equation supplies more information than finite invertibility: it specifies the single trajectory to be inverted. Its uniqueness proof divides only on a safe transform line and supplies no uniform bound for an inverse of convolution by  $\mathcal{A}$ . Its nonzero limiting forcing motivates the centering in the next section.

## 5. Centering and cancellation

All moments used to center the forcing are ordinary convergent integrals. From the preceding section,

$$U_* = \int_0^\infty g(t) dt = G(0) = \frac{4}{3L_0(0)} > 0, \quad (75)$$

$$C_* = \int_0^\infty v(t) dt = \left(\frac{\pi}{3}\right)U_*, \quad c = 6C_* = 2\pi U_* = 8\frac{\pi}{3L_0(0)}. \quad (76)$$

Here  $L_0(0) = 2 + \frac{\gamma}{2} - \frac{\log(4\pi)}{2} > 0$ . Define the causal transient

$$\mathcal{R}(t) = \mathcal{H}(t) - c = v'(t) + 5v(t) - 6 \int_t^\infty v(s) ds. \quad (77)$$

Subtraction of the constant means the constant on positive time, not a whole-line distribution. As  $\int \mathcal{A} = L_0(0)$ , (73) yields the centered equation

$$\mathcal{A} * \mathcal{R} = W - 8\frac{\pi}{3} + c \int_t^\infty \mathcal{A}(s) ds. \quad (78)$$

Its right side has a PNT-scale stretched-exponential envelope. It is independently specified by the prime discrepancy and the  $\Gamma$  kernel.

The onset formula gives  $u(0+) = g(0+) = 0$ . The explicit kernel  $K_0$  is locally of bounded variation with  $K_0(0+) = 2\pi$ ; hence  $v = K_0 * g$  has  $v'(0+) = 2\pi g(0+) = 0$ . Thus  $\mathcal{H}(0+) = 0$ . The finite identity (60) gives, at every finite time with the natural continuous convention,

$$2\pi g(t) = cb_\mu(e^t) + (Q_e \mathcal{R})(t), \quad b_\mu(x) = \sum_{n \leq x} \frac{\mu(n)}{n}. \quad (79)$$

The  $n = 1$  term is included. At a new integer the apparent jumps cancel because  $\mathcal{H}(0) = 0$  and  $\mathcal{R}(0) = -c$ . On an  $L^2$  window the last endpoint has no effect.

In  $\Re z > 0$ , all following transforms are ordinary and absolute:

$$\widehat{b_\mu(e)}(z) = \frac{1}{z\zeta(1+z)}, \quad (80)$$

$$\hat{\mathcal{R}}(z) = 2\pi\zeta(1+z)G(z) - \frac{c}{z}. \quad (81)$$

Consequently the assembled transform is

$$\frac{c}{z\zeta(1+z)} + \frac{\hat{\mathcal{R}}(z)}{\zeta(1+z)} = 2\pi G(z). \quad (82)$$

This identity describes cancellation between the constant and transient terms before either is estimated.

It also distinguishes  $\zeta$  zeros from genuine response poles. At a zero  $\rho$  of  $\xi$  of order  $m$ , writing  $s = \rho + w$  gives

$$\frac{\xi(s)}{\xi(s) + \xi'(s)} = \frac{w}{m} + O(w^2). \quad (83)$$

Thus the reduced  $G$  has a simple zero at  $z = \rho - 1$ ;  $\hat{\mathcal{H}} = 2\pi\zeta(1+z)G(z)$  vanishes there to order  $m + 1$ . The apparent  $\frac{1}{\zeta}$  singularities in (82) cancel. This is an identity of analytic continuations at those points, not an assertion that the left-half-plane moments converge. Zeros of  $\xi + \xi'$  at which  $\xi \neq 0$  remain the genuine possible poles.

The next section proves that  $\mathcal{R}$  has all polynomially weighted  $L^2$  norms and a square-root-exponential weight. In particular  $\mathcal{H}(T + \cdot) \rightarrow c$  in  $L^2$  on every fixed bounded interval, with a stretched-exponential rate. This is the limiting topology used here. A pointwise derivative limit for  $v'$  is not required.

## 6. Weighted moments and stretched-exponential control

The first derivative estimate (67) does not by itself iterate to all orders: a second derivative contains an unabsorbed factor of  $z$  multiplying products of  $\hat{E}'_0$ . We obtain uniform estimates instead from logarithmic complex neighborhoods of the actual response.

**Theorem 5** (Gevrey-2 time moments).

There are constants  $A, B > 0$  such that, for every integer  $k \geq 0$  and every  $q \in \{g, v, v', \mathcal{R}\}$ ,

$$\|t^k q\|_2 \leq AB^k (k!)^2. \quad (84)$$

Consequently, for some  $a_0 > 0$ ,  $e^{a_0\sqrt{t}}q$  belongs to  $L^2(0, \infty)$  and, after reducing  $a_0$ , to  $L^1(0, \infty)$ . The two factorials come from different operations. A logarithmic zero-free region supplies complex disks of radius comparable to  $\frac{1}{\log(2+|T|)}$ . Cauchy's estimate on such a disk produces  $k!$  and a logarithmic power. Integrating the squared bound vertically produces  $(2k)!$ ; its square root has another factorial-scale bound. Plancherel then converts transform derivatives into time moments, and choosing the moment order proportional to  $\sqrt{R}$  gives the tail estimate. This explains the order proved here without asserting that it is optimal for the literal response.

**Proof. Zero-free disks.** Let  $\rho = \beta_\rho + i\gamma_\rho$  range over zeros of  $\xi$ , with multiplicity. The paired product gives the absolutely convergent real-part identity

$$\Re\left(\frac{\xi'}{\xi}\right)(x + iT) = \sum_\rho \frac{x - \beta_\rho}{(x - \beta_\rho)^2 + (T - \gamma_\rho)^2}. \quad (85)$$

It follows either by taking real parts in the symmetric product for  $\xi(\frac{1}{2} + r)$  or by pairing the usual order-one Hadamard factors; the functional equation cancels the real constant. At  $x = 2$ , since  $0 \leq \beta_\rho \leq 1$ ,

$$D(T) := \sum_{\rho} \frac{1}{1 + (T - \gamma_\rho)^2} \leq 4\Re\left(\frac{\xi'}{\xi}\right)(2 + iT) \leq 64 \log(2 + |T|). \quad (86)$$

For the last bound use

$$\left(\frac{\xi'}{\xi}\right)(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{\log(\pi)}{2} + \frac{\psi(\frac{s}{2})}{2} + \frac{\zeta'(s)}{\zeta(s)}. \quad (87)$$

On  $\Re s = 2$  the Euler series bounds  $|\frac{\zeta'}{\zeta}|$  by  $\sum_{n \geq 2} \log(n)n^{-2} \leq 3$ . Splitting the digamma series at  $n$  comparable to  $|T|$  bounds it by  $\log(2 + |T|) + 6$ . These loose constants suffice.

The classical explicit zero-free region [7, Theorem 1] gives

$$\zeta(s) \neq 0 \quad \text{if} \quad |\Im s| \geq 2, \quad \Re s > 1 - \frac{1}{5.573412 \log |\Im s|}. \quad (88)$$

Put  $l(T) = \log(2 + |T|)$  and  $\delta_0 = \frac{1}{2048}$ . When  $|T| \geq 4$ , consider the disk  $|z - iT| \leq \frac{\delta_0}{l(T)}$ . The zeros with  $|\gamma_\rho - \Im z| \leq 1$  have  $2 \leq |\gamma_\rho| \leq |T| + 2$ ; (88) makes their contributions to (85), at  $s = 1 + z$ , nonnegative. Any negative contribution from the remaining zeros has magnitude at most  $|\Re z| \frac{1}{(\Im z - \gamma_\rho)^2}$ . By (86) their total is at most  $2|\Re z|D(\Im z) \leq 256\delta_0 \leq \frac{1}{8}$ . No zero of  $\xi(1 + z)$  lies in the disk, by the same near-height exclusion. Thus  $\Re L_0(z) \geq \frac{1}{2}$  there.

On bounded heights, nonvanishing on  $\Re s = 1$ , which follows from the prime number theorem, and (85) give a fixed tubular neighborhood where  $L_0$  is holomorphic and  $\Re L_0 \geq \frac{1}{2}$ . Choose a positive  $\rho_c \leq \frac{1}{4}$  for that compact neighborhood and put

$$\delta = \min\left(\frac{1}{2048}, \rho_c \frac{\log(2)}{2}\right) > 0. \quad (89)$$

Then on every disk  $|z - iT| \leq \frac{\delta}{l(T)}$ ,

$$|G(z)| \leq \frac{12}{1 + |T|}. \quad (90)$$

**Cauchy estimates and time moments.** Cauchy's estimate yields

$$|G^{(k)}(iT)| \leq 12k! \delta^{-k} \frac{l(T)^k}{1 + |T|}. \quad (91)$$

The elementary integral estimate

$$\int_0^\infty \frac{l(T)^{2k}}{(1 + T)^2} dT \leq 4(2k)! \quad (92)$$

follows by  $u = \log(2 + T)$ . Differentiation of the already identified Fourier boundary in the sense of distributions, followed by Plancherel, proves

$$\|t^k g\|_2 \leq \frac{24}{\sqrt{\pi}} \left(\frac{2}{\delta}\right)^k (k!)^2, \quad (93)$$

using  $\sqrt{(2k)!} \leq 2^k k!$ . This proves the time moments rather than assuming them in the differentiation.

**The output and centered transient.** The other channels have the same order. On these disks,

$$|m_0(z)| \leq 64\pi \frac{l(T)}{1 + |T|}. \quad (94)$$

For  $|T| \geq 4$ , one-step Euler summation [9, (25.2.8)] for  $\zeta(1+z)$  with cutoff  $\lceil |T| + 2 \rceil$  bounds its finite sum, polar integral and remainder by  $1 + 2\log(|T| + 3)$ ,  $\frac{2}{|T|}$  and an absolute constant, respectively. Here  $|\Re z| \log(|T| + 3)$  is uniformly small. Multiplying by the rational factor in  $m_0$  proves (94). On bounded heights use the convergent kernel estimate  $|K_0(t)| \leq 2\pi e^{-t}$  in a neighborhood with  $\Re z \geq -\frac{1}{4}$ .

Both  $V = m_0 G$  and  $zV$ , the transform of  $v'$ , consequently have a disk bound of the form  $C \frac{l(T)}{1+|T|}$ . There is no initial derivative atom since  $v(0) = 0$ . Repeating (91) and Plancherel, and absorbing  $k + 1$  into a geometric constant, gives a common estimate

$$\|t^k q\|_2 \leq A_G B_G^k (k!)^2, \quad q \in \{g, v, v'\}, \quad (95)$$

for example with  $A_G = 48 \frac{64\pi}{\sqrt{\pi}}$  and  $B_G = \frac{4}{\delta}$ . For the transient use the weighted Hardy tail estimate

$$\left\| t^k \int_t^\infty v(s) ds \right\|_2 \leq \frac{2}{2k+1} \|t^{k+1} v\|_2. \quad (96)$$

To prove it, integrate the squared tail against the derivative of  $t^{2k+1}$ , then apply Cauchy–Schwarz; truncate first and use the already established higher moments for the endpoints. Equation (77) now gives (84), for instance with  $A = A_G(6 + 12B_G)$ ,  $B = 2B_G$ . The compact-height radius is positive but not numerically evaluated.

**Optimizing the moment order.** Choose  $k = \lfloor \sqrt{\frac{R}{4B}} \rfloor$ . Since  $k! \leq k^k$ ,

$$\|1_{t \geq R} q\|_2 \leq 4A e^{-\kappa_0 \sqrt{R}}, \quad \kappa_0 = \frac{\log(2)}{\sqrt{B}}. \quad (97)$$

Layer-cake integration gives the square-root-exponential  $L^2$  weight for every  $a_0 < \kappa_0$ . Cauchy–Schwarz with a slightly stronger weight gives the asserted  $L^1$  property.  $\square$

The zero-free input in (88) is used in shrinking disks, not a fixed left half-strip. The theorem implies

$$\|\mathcal{H} - c\|_{L^2(T, T+1)} \leq 4A e^{-\kappa_0 \sqrt{T}}. \quad (98)$$

Together with (79) it bounds the sum  $cb_\mu + Q_e \mathcal{R}$ , retaining its cancellation. For  $0 < \sigma < \frac{1}{2}$ , re-conjugation still allows a bound of the form  $O(e^{(\frac{1}{2}-\sigma)R - \kappa' \sqrt{R}})$  on the physical cutoff norm. The positive linear exponent has not been removed.

The moment estimate itself cannot imply such removal. A smooth modification near zero of  $q_a(t) = e^{-a\sqrt{t}}$  has

$$\|t^k q_a\|_2^2 = 2 \frac{\Gamma(4k+2)}{(2a)^{4k+2}} \quad (99)$$

up to the harmless local modification, and satisfies a Gevrey-2 bound. Yet  $e^{\varepsilon t} q_a$  is not in  $L^2$  for any  $\varepsilon > 0$ . This is a limitation of the moment scale, not an optimality assertion about the literal response.

## 7. The fractional-range threshold

We restate the physical operators needed from the causal-range criterion. With  $p$  normalized by  $M$ , let  $a(t) = (1 - t)p(t)$ ,  $b(t) = (1 + t)p(t)$ , and on  $L^2(0, \infty)$  set

$$H_a f(t) = \int_0^\infty a(t+s)f(s) ds, \quad H_b f(t) = \int_0^\infty b(t+s)f(s) ds, \quad b_* = H_a \kappa. \quad (100)$$

The operators are compact self-adjoint and injective. For example,  $H_b f = 0$  makes  $\int b(z+s)f(s) ds$  vanish throughout the  $\theta$  strip by analyticity, after which full-line Fourier uniqueness gives  $f = 0$ .

**Proposition (The ordinary causal-preimage endpoint).** The distinguished source satisfies

$$b_* \in \text{Ran } H_b \Leftrightarrow q_c \in L^2(0, \infty). \quad (101)$$

When these conditions hold,  $q_c$  is the unique ordinary Hankel preimage and

$$\|q_c\|_2^2 = \frac{1}{2}. \quad (102)$$

**Proof.** The full-line equation  $a * y = \kappa * b$  has a unique ordinary  $L^2$  solution of squared norm  $\frac{1}{2}$ , by its unimodular Fourier multiplier [1, Theorem “Range and causality”]. Its one-sided support is not assumed. If  $H_b q = b_*$  for ordinary  $q \in L^2(0, \infty)$ , extend  $q$  by zero. The residual  $a * q - \kappa * b$  is holomorphic in the  $\theta$  strip by uniform tail domination, and evaluating it at negative time gives zero. Analytic continuation makes it vanish on the full line, so Fourier uniqueness identifies the extension with  $y$ . It also lies in the weighted class of Section 2; uniqueness on that common safe transform line identifies it with  $q_c$ . This proves the norm and uniqueness assertions. Conversely, if  $q_c$  is ordinary  $L^2$ , evaluating (25) at negative time gives the Hankel equation.  $\square$

Even before ordinary square integrability is known, this last Hankel integral for  $q_c$  converges absolutely against the double-exponential kernel tail. That extended integral identity is distinct from membership in the actual operator range. The endpoint proposition uses neither range closure nor a bounded inverse of  $H_b$ .

For orientation, if  $T = |H_b|$  has eigenvalues  $t_n > 0$  and orthonormal eigenvectors  $e_n$ , the spectral range criterion is

$$b_* \in \text{Ran } T^p \Leftrightarrow \sum_n \frac{|\langle e_n, b_* \rangle|^2}{t_n^{2p}} < \infty. \quad (103)$$

The threshold theorem controls these weighted coefficients from two different directions. Below the first power, truncated causal solutions give approximate preimages: their norms grow exponentially with the truncation length, but their Hankel errors decay doubly exponentially. Above the first power, a preimage would gain smoothness at the initial endpoint, contradicting the logarithmic onset of  $q_c$ . The first mechanism concerns approximation over long intervals; the second concerns local boundary regularity.

For the original Hankel operator, put  $A_H = H_b^2$  and

$$\mathcal{E}(\lambda) = \langle b_*, ((A_H + \lambda I)^{-1} b_*) \rangle, \quad \lambda > 0. \quad (104)$$

**Proposition (Regularized inverse energy).** There is a constant  $C$ , independent of  $\lambda$ , such that

$$\mathcal{E}(\lambda) \leq C \left[ 1 + \log \left( \frac{1}{\lambda} \right) \right]^{\frac{1}{2}}, \quad 0 < \lambda < 1. \quad (105)$$

**Proof. Approximate preimages.** Evaluate (25) at negative time. Its integral converges against the double-exponential Hankel tail, even before  $q_c$  is known to be unweighted square-integrable. Hence  $H_b q_c = b_*$  in this extended integral sense. Let  $q_R = \mathbf{1}_{(0,R)} q_c$ .

The edge response  $g = e^{-\frac{t}{2}}(q_c + \kappa)$  belongs to  $L^2$ , so

$$C_q := \|e^{-\frac{t}{2}} q_c\|_2 < \infty, \quad \|q_R\|_2 \leq C_q e^{\frac{R}{2}}. \quad (106)$$

For the original kernel  $b = (1+t)p(t)$ , weighted Cauchy–Schwarz and Tonelli give

$$\begin{aligned} \|H_b q_R - b_*\|_2^2 &\leq C_q^2 \int_0^\infty \int_R^\infty e^u |b(t+u)|^2 du dt \\ &= C_q^2 \int_R^\infty (e^v - e^R) |b(v)|^2 dv. \end{aligned} \quad (107)$$

The actual  $\theta$  series implies  $|p(v)| \leq C_\theta e^{9\frac{v}{2} - \pi e^{2v}}$  for  $v \geq 0$ . To see this, factor out the  $n = 1$  exponential; the remaining polynomial coefficients are bounded by a constant times the summable series  $\sum_{n \geq 1} n^4 e^{-\pi(n^2-1)}$ . For  $R \geq 1$ , the logarithmic derivative of  $(1+v)^2 e^{10v-2\pi e^{2v}}$  is at most  $12 - 4\pi e^{2R} < 0$  on the tail. Hence

$$\|H_b q_R - b_*\|_2^2 \leq C C_q^2 (1+R)^2 \frac{e^{10R-2\pi e^{2R}}}{4\pi e^{2R} - 12}. \quad (108)$$

For the actual decomposition  $b_* = H_b q_R + d_R$ , functional calculus gives

$$\mathcal{E}(\lambda)^{\frac{1}{2}} \leq \|q_R\|_2 + \frac{\|d_R\|_2}{\sqrt{\lambda}}, \quad (109)$$

because  $\|(A_H + \lambda I)^{-\frac{1}{2}} H_b\| \leq 1$ . Write  $T = \log(\frac{1}{\lambda})$  and choose

$$R = \frac{1}{2} \log \left( \frac{T}{\pi} \right), \quad T \geq \pi e^2. \quad (110)$$

The first term in (109) is  $O(T^{\frac{1}{4}})$ ; (108) makes the second  $O((1 + \log T) T^2 e^{-\frac{T}{2}})$ . Squaring after these estimates proves (105). Enlarging  $C$  handles the remaining compact parameter interval, using  $\mathcal{E}(\lambda) \leq \frac{\|b_*\|_2^2}{\lambda}$ .  $\square$

**Theorem 6** (Fractional ranges of the distinguished source).

For the literal  $\theta$  operators,

$$b_* \in \text{Ran}|H_b|^{1-\delta} \quad (0 < \delta < 1), \quad (111)$$

$$b_* \notin \text{Ran}|H_b|^{1+\delta} \quad (\delta > 0). \quad (112)$$

The first-power range condition is left undecided.

**Proof. Subcritical ranges.** If  $\mu_*$  is the spectral measure of  $b_*$  for  $A_H$ , then for  $0 < \delta < 1$  the scalar identity

$$t^{-1+\delta} = \frac{\sin(\pi\delta)}{\pi} \int_0^\infty \frac{\lambda^{\delta-1}}{t+\lambda} d\lambda \quad (113)$$

and Tonelli show  $\int t^{-1+\delta} d\mu_*(t) < \infty$ . Near zero this follows from (105); at infinity use  $\mathcal{E}(\lambda) \leq \frac{\|b_*\|^2}{\lambda}$ . The spectral range criterion proves the subcritical range assertion.

**Extra smoothing and the initial boundary.** For the second line, every positive power of  $|H_b|$  maps  $L^2$  into every finite Sobolev space on the half-line. First  $H_b : L^2 \rightarrow H^m$  is bounded for every integer  $m$ , because  $\int_0^\infty u|b^{(m)}(u)|^2 du < \infty$ . The identity  $|H_b| = H_b \operatorname{sgn}(H_b)$  proves the same for  $|H_b|$ . Normalize this positive operator as  $T$  with  $\|T\| \leq 1$ , and decompose  $f$  into spectral blocks  $f_j$  where  $2^{-j-1} < T \leq 2^{-j}$ . For  $0 < \delta < 1$  and  $u_j = T^\delta f_j$ ,

$$\|u_j\|_2 \leq 2^{-j\delta} \|f_j\|, \quad \|u_j\|_{H^m} \leq C_m 2^{j(1-\delta)} \|f_j\|. \quad (114)$$

Half-line Sobolev interpolation gives  $\|u_j\|_{H^s} \leq C 2^{j(\frac{s}{m}-\delta)} \|f_j\|$ . Choose  $m > \frac{s}{\delta}$  and sum by Cauchy–Schwarz. Larger powers factor off one smoothing power. Sobolev embedding therefore gives smoothness up to zero.

If  $b_* = |H_b|^{1+\delta} f$ , then

$$q = \operatorname{sgn}(H_b) |H_b|^\delta f = |H_b|^\delta \operatorname{sgn}(H_b) f \quad (115)$$

is an ordinary Hankel preimage and is smooth at zero. The commutation here is spectral calculus; no Sobolev preservation by  $\operatorname{sgn}(H_b)$  is assumed. The range criterion identifies  $q$  with  $q_c$ . This contradicts the logarithmic onset (44). Equivalently, (21) and the real digamma asymptotic give  $R\hat{q}_c(R) + 1 \sim \frac{4}{\log(R)}$ , while any smooth-at-zero  $L^2$  preimage would have  $R\hat{q}(R) = q(0) + O(R^{-1})$ . Thus no supercritical power is possible.  $\square$

**Corollary (Weighted inverse-energy bounds).** There are constants  $C, c > 0$ , independent of  $\lambda$ , such that, with  $T = \log(\frac{1}{\lambda})$ ,

$$\mathcal{E}(\lambda) \leq C \left[ 1 + \sqrt{T} e^{-c\sqrt{\log T}} \right], \quad T \geq \max(\pi e^2, \pi^2). \quad (116)$$

If in addition  $e^{\beta t} g \in L^2$  for some  $0 < \beta < \frac{1}{2}$ , then

$$\mathcal{E}(\lambda) = O_\beta \left( \left[ \log\left(\frac{1}{\lambda}\right) \right]^{\frac{1}{2}-\beta} \right) \quad (\lambda \downarrow 0). \quad (117)$$

**Proof.** The Gevrey theorem in Section 6 supplies  $a > 0$  for which  $e^{a\sqrt{t}} g \in L^2$ . Since  $q_c = e^{\frac{t}{2}} g - \kappa$ , convexity of  $\frac{t}{2} - a\sqrt{t}$  as a function of  $\sqrt{t}$  gives

$$\|q_R\|_2 \leq C \left[ 1 + e^{\frac{R}{2} - a\sqrt{R}} \right]. \quad (118)$$

The maximum on  $0 \leq t \leq R$  occurs at an endpoint; the constant term accounts for the initial endpoint. Use the same original-kernel residual (108) and  $R = \frac{\log(\frac{T}{\pi})}{2}$ . After squaring (109), the main term is bounded by

$$C \left[ 1 + \sqrt{\frac{T}{\pi}} e^{-\sqrt{2}a\sqrt{\log(\frac{T}{\pi})}} \right]. \quad (119)$$

For the stated threshold,  $\log(\frac{T}{\pi}) \geq \frac{\log(T)}{2}$ , so (116) follows, for example with  $c = a$ . The exponentially small residual term is absorbed by the constant.

Under the additional exponential-weight hypothesis, the same argument instead gives

$$\|q_R\|_2 \leq C_\beta \left[1 + e^{(\frac{1}{2}-\beta)R}\right]. \quad (120)$$

Squaring at the same cutoff yields (117). This is a conditional weight estimate, with constants allowed to depend on  $\beta$ .  $\square$

The quantity  $\mathcal{E}(\lambda)$  in (105) is the regularized inverse energy of the causal-range companion [1, Theorem “Inverse energy and rank-one detection”]. Its limit is  $\frac{1}{2}$  when an ordinary preimage exists and infinity otherwise. The square-root-logarithmic bound and its Gevrey refinement control the possible growth of this original inverse energy, while allowing either limit. The kernel transport in Section 2.1 does not identify this positive-parameter energy with that of the transported operator.

This threshold concerns compact-operator smoothing and the short-time boundary of a particular source. The Gevrey theorem concerns its long-time response. Neither substitutes one endpoint for the other.

## 8. A fixed pole gap and a smaller Hardy shift

A pole of the reduced response (18) is called genuine if its denominator zero is not cancelled by its numerator. At a zero of  $M$  of order  $m$ ,  $\frac{M}{M+M'}$  has a simple zero, so such a point is harmless for the response. The following theorem assumes a gap only for the genuine poles.

**Theorem 7** (Conditional exponential return).

Suppose, for some  $0 \leq a < \frac{1}{2}$ , the reduced  $U$  has no genuine pole in  $\Re r > a$ . Put

$$\beta_a = \frac{\left(\frac{1}{2} - a\right)^2}{16(2-a)}. \quad (121)$$

Then  $U \in H^2(\Re r > \frac{1}{2} - \beta_a)$ , or equivalently

$$e^{\beta_a t} g(t) \in L^2(0, \infty). \quad (122)$$

The Hardy constant is finite and depends on  $a$  and the literal completed pair.

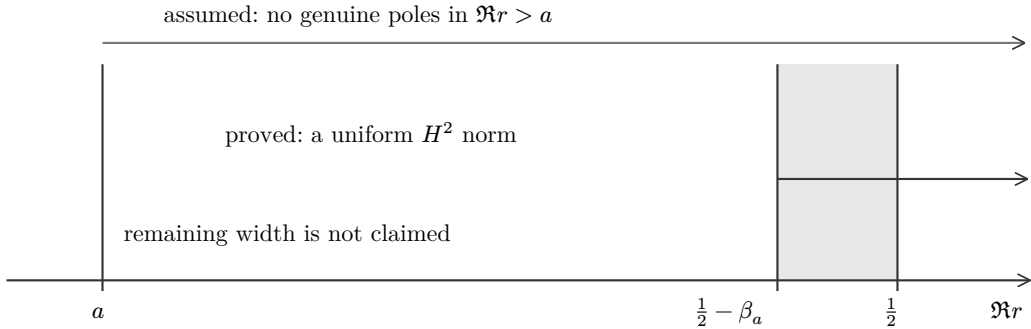


Figure 1: The conditional width loss, drawn schematically. A genuine-pole gap to  $a$  yields the smaller shift  $\beta_a = \frac{(\frac{1}{2}-a)^2}{16(2-a)}$ . The shaded band is the gain beyond the unconditional line  $\Re r = \frac{1}{2}$ ; the Hardy estimate holds to its right.

The pole gap gives holomorphy in one half-plane; the proof still has to control vertical growth. Its representation changes are

common-factor cancellation  $\rightarrow$  zero-free denominator  $\rightarrow$  Harnack bound  
 $\rightarrow$  three-lines interpolation  $\rightarrow$  smaller Hardy half-plane

After removing common factors, the reciprocal of the remaining denominator is controlled on a safe line. Harnack transports that bound across the assumed pole-free region. Interpolation with the stronger Euler-edge estimate then gives integrable vertical decay close enough to that edge. The final boundary is  $\frac{1}{2} - \beta_a$ , strictly to the right of the assumed boundary  $a$ .

**Proof. The completed pair and its common factors.** We first give the factorization used in the proof. Define

$$A_+(r) = m(r) + \dot{m}(r), \quad A_-(r) = m(r) - \dot{m}(r), \quad (123)$$

where the dot is the parameter derivative of

$$m_\vartheta(r) = 2 \frac{\xi(\vartheta + r)}{(r + \frac{5}{2})(r + \frac{3}{2})\Gamma_{\mathbb{R}}(r + \frac{5}{2})} \quad (124)$$

at  $\vartheta = \frac{1}{2}$ . It is not the derivative in  $r$ . The two functions belong to  $H^\infty(\Re r > 0)$ . Here is an analytic bound sufficient for this fact. Expanding  $\xi'$  into  $\zeta$  and  $\zeta'$  and using (48) gives, with  $s_\zeta = r + \frac{1}{2}$  and  $R_0(r) = 2 \frac{\pi(r - \frac{1}{2})}{(r + \frac{3}{2})(r + \frac{5}{2})}$ ,

$$\dot{m}(r) = R_0(r) \left\{ \zeta'(s_\zeta) + \left[ \frac{1}{s_\zeta} + \frac{1}{s_\zeta - 1} - \frac{\log(\pi)}{2} + \frac{\psi(\frac{s_\zeta}{2})}{2} \right] \zeta(s_\zeta) \right\}. \quad (125)$$

Apparent singularities are removable in the original completed expression. One-step Euler summation with cutoff comparable to  $2 + |\Im r|$  gives  $\zeta(s_\zeta) = O(|\Im r|^{\frac{1}{2}})$  and  $\zeta'(s_\zeta) = O(|\Im r|^{\frac{1}{2}} \log(2 + |\Im r|))$  uniformly on  $0 \leq \Re r \leq 2$  at large height. The digamma bound is logarithmic, so (125) and (48) are bounded there. On  $\Re r \geq 2$  absolute Euler convergence and the same  $\Gamma$  asymptotic give boundedness. Compact sets are covered by analytic continuation.

On the imaginary boundary, the functional equation makes  $M(iT)$  real and  $M'(iT)$  imaginary. Hence  $|A_+(iT)| = |A_-(iT)|$ . We use the scalar factorization theorem in the following form: a nonzero bounded analytic function on a half-plane is an outer factor times a Blaschke product, a singular inner factor on the finite boundary, and a possible exponential inner factor at infinity. The boundary modulus determines the outer factor up to a unimodular constant [10, Introduction and Corollary 5.11, classical one-variable case]. Thus the equal boundary moduli give

$$A_+ = OB_+, \quad A_- = OB_-, \quad (126)$$

with the same bounded outer factor  $O$  (unimodular constants absorbed) and inner factors. The inner factors are pure Blaschke products. Indeed the explicit completed formulas continue across every finite boundary point with isolated finite-order zeros. A singular measure can be supported only on that discrete zero set; an atom would impose normal decay  $e^{-\frac{c}{x}}$  incompatible with a finite Taylor order and a bounded remaining factor. There is no infinity delay either, since on the positive real axis

$$|A_+(x)| \sim \pi \frac{\log(x)}{x}, \quad |A_-(x)| \sim \pi \frac{\log(x)}{x}. \quad (127)$$

These asymptotics follow from (48), the Euler series and the digamma expansion, and exclude an exponential inner factor. No inverse of the outer factor has been bounded.

Remove their common Blaschke divisor:  $B_+ = CD_0$ ,  $B_- = CN_0$ , with  $D_0, N_0$  coprime. Then

$$S_0 = \frac{M - M'}{M + M'} = \frac{N_0}{D_0}, \quad F(r) := (r + 1)U(r) = 1 + S_0(r). \quad (128)$$

Zeros of  $D_0$  are precisely genuine response poles in the right half-plane. The hypothesis makes  $D_0$  zero-free on  $\Re r > a$ .

**A reciprocal bound from the gap.** On  $r = 2 + iT$ , the paired product gives  $\Re\left(\frac{M'}{M}\right) \geq 0$ . The reciprocal Euler series for  $\zeta\left(\frac{5}{2} + iT\right)$  and the rational formula for  $m$  give  $|m(2 + iT)| \geq \frac{c}{1+|T|}$ . Consequently

$$|D_0(2 + iT)| \geq |B_+(2 + iT)| \geq |A_+(2 + iT)| \frac{1}{\|O\|_\infty} \geq \frac{c_1}{1 + |T|}. \quad (129)$$

The first inequality uses  $|C| \leq 1$ . The function  $h = -\log|D_0|$  is nonnegative harmonic on  $\Re r > a$ . The half-plane Harnack inequality, along a horizontal segment at height  $T$ , gives

$$h(x + iT) \leq \frac{2-a}{x-a} h(2 + iT), \quad a < x \leq 2. \quad (130)$$

For example map the half-plane to the disk taking  $2 + iT$  to zero; its radial Harnack factor is exactly  $\frac{2-a}{x-a}$ .

**Interpolation and the Hardy norm.** Let  $d = \frac{1}{2} - a$ ,  $x_0 = \frac{1}{2} - \frac{d}{2}$  and  $K_a = 2^{\frac{2-a}{d}}$ . For  $x_0 \leq x \leq \frac{1}{2}$ , (130) gives  $|F(x + iT)| \leq C_a(1 + |T|)^{K_a}$ . On the right edge (62) gives the stronger  $|F(\frac{1}{2} + iT)| \leq 2$ . For fixed  $T$ , apply the bounded three-lines theorem to  $F(r)e^{(r-iT)^2}$  on this strip. Gaussian decay controls the imaginary ends, and the left and right boundary suprema are respectively  $C_a(1 + |T|)^{K_a}$  and a constant. Hence

$$|F(x + iT)| \leq C_a(1 + |T|)^{K_a \frac{\frac{1}{2}-x}{\frac{1}{2}}}. \quad (131)$$

When  $x \geq \frac{1}{2} - \beta_a$ , this exponent is at most  $\frac{1}{4}$ . Therefore  $|U(x + iT)| \leq C_a(1 + |T|)^{-\frac{3}{4}}$  uniformly throughout the smaller closed strip. On  $\Re r \geq \frac{1}{2}$  the bound  $\frac{2}{|r+1|}$  supplies the rest of the uniform vertical  $L^2$  norm. Thus  $U$  belongs to the stated Hardy space. Laplace–Hardy inversion and uniqueness on a common safe line identify its inverse with the pre-existing causal  $u$ , proving (122).  $\square$

The proof loses width: it does not assert the Hardy norm at the boundary  $\Re r = a$ . For the literal derivative pair, however, it removes a separate growth assumption when the question is whether some positive exponential weight exists.

## 9. The remaining endpoint

The causal response is defined without spectral data, and every finite physical interval has the finite arithmetic reconstruction (19) and (54). Its edge response has the unconditional regularity (84). The compact Hankel source reaches every power below the first and no power above it. The first-power question remains whether the supported response is the ordinary square-integrable inverse, as specified by (101). The square-root-logarithmic bound (105) and its Gevrey refinement (116) control the possible divergence of the original regularized inverse energy without deciding that endpoint.

The conditional theorem yields the precise existential equivalence

$$\left( \exists a \in \left[0, \frac{1}{2}\right) : U \text{ has no genuine pole in } \Re r > a \right) \Leftrightarrow (\exists \beta > 0 : e^{\beta t} g \in L^2). \quad (132)$$

The reverse implication is Hardy holomorphy, reducing  $\beta$  if necessary. This is a fixed-gap statement; the proof supplies no such gap unconditionally.

For clarity, the all-weight endpoint is stronger. If all genuine poles are absent in  $\Re r > 0$ , the denominator Blaschke factor  $D_0$  in Section 8 is constant. Thus  $S_0 = \frac{N_0}{D_0}$  is a Schur function.

It is nonconstant, since  $S_0(0) = 1$  by continuation whereas  $S_0(x) \rightarrow -1$  on the positive real axis. Any interior zero of  $M$ , simple or multiple, would give the removable value  $S_0 = -1$ , contradicting the maximum modulus principle. The functional equation then confines all nontrivial  $\xi$  zeros to the critical line. Conversely, under that zero-location assertion the paired product gives  $\Re\left(\frac{M'}{M}\right) \geq 0$  on  $\Re r > 0$ , hence  $|U(r)| \leq \frac{2}{|r+1|}$  there. This bound also reaches the ordinary endpoint, because

$$\int_{\mathbb{R}} |U(\sigma + iT)|^2 dT \leq 4 \frac{\pi}{1 + \sigma} \leq 4\pi, \quad \sigma > 0. \quad (133)$$

Thus  $U$  belongs to the Hardy space of the full right half-plane with a uniform vertical norm. Laplace–Hardy inversion gives an ordinary  $L^2$  causal inverse, and uniqueness on the original safe transform line identifies it with  $u$ . Consequently  $q_c = u - \kappa \in L^2$ , which is the actual Hankel-range condition of Section 7. Since  $u = e^{\frac{t}{2}}g$ , this also gives every weight  $e^{\beta t}g \in L^2$ ,  $0 < \beta < \frac{1}{2}$ . The endpoint follows from (133) for this literal response, rather than by taking  $\beta$  to  $\frac{1}{2}$  in separate weighted estimates. Therefore

$$\begin{aligned} \text{RH} &\Leftrightarrow U \text{ has no genuine pole in } \Re r > 0 \\ &\Leftrightarrow e^{\beta t}g \in L^2 \text{ for every } 0 < \beta < \frac{1}{2}. \end{aligned} \quad (134)$$

The ordinary endpoint itself belongs to the same list.

**Corollary (The range endpoint and RH).** For the literal  $\theta$  operators and the fixed source normalization,

$$\text{RH} \Leftrightarrow q_c \in L^2(0, \infty) \Leftrightarrow b_* \in \text{Ran } H_b. \quad (135)$$

**Proof.** The second equivalence is (101). If RH holds, the uniform Hardy bound (133) gives  $q_c \in L^2$ , as shown above. Conversely, suppose  $q_c \in L^2(0, \infty)$ . Then  $u = q_c + \kappa \in L^2(0, \infty)$ , so its Laplace transform  $\hat{u}$  is holomorphic on  $\Re r > 0$ . By (18), the identity

$$(r+1)(M(r) + M'(r))\hat{u}(r) = 2M(r) \quad (136)$$

holds on the initial half-plane  $\Re r \geq 4$ , and therefore on all of  $\Re r > 0$  by analytic continuation. At a zero of  $M + M'$  of order  $m$  in this half-plane, the left side vanishes to order at least  $m$ , and hence so does  $M$ . The quotient  $U$  is therefore holomorphic there. Thus  $U$  has no genuine pole in  $\Re r > 0$ , and (134) gives RH.  $\square$

Combined with [1, Theorems “Correlation realization” and “Range and causality”], RH is therefore also equivalent to positive semidefiniteness of every finite positive-rate matrix. The corollary identifies the strength of the range condition; it does not decide it.

This derivative-pair endpoint is consistent with the classical half-plane criteria of Lagarias and the related Hilbert-space formulations of Suzuki. The quantitative theorem of Section 8 concerns the smaller Hardy shift obtained from one assumed fixed gap.

The square-root-exponential estimates proved here neither verify that gap nor provide a counterexample to it. They give unconditional source estimates up to a sharply specified range and analytic boundary, while preserving the literal arithmetic reconstruction on every finite interval.

## References

- [1] Suvlu, Dylan,  $\theta$  correlations and a causal range criterion for the Weil form, 2026. Companion manuscript, version v0.1.0-draft.4, revised 24 September 2026.
- [2] Lagarias, Jeffrey C., Zero spacing distributions for differenced L-functions, *Acta Arithmetica* **120** (2005) 159–184. Section 2; preprint version 2, 30 January 2006.
- [3] Lagarias, Jeffrey C., Hilbert spaces of entire functions and Dirichlet L-functions, in *Frontiers in Number Theory, Physics, and Geometry I*, Springer, 2006, pp. 365–377. Theorem 1; author preprint, 2 January 2006, pp. 9–10.
- [4] Suzuki, Masatoshi, On the Hilbert space derived from the Weil distribution, *Canadian Journal of Mathematics* (2025) 1–24, doi:10.4153/S0008414X25101739. Online publication; arXiv version 3, 7 November 2025.
- [5] Suvlu, Dylan, An operator-theoretic representation of the Weil quadratic form, 2026. Companion manuscript, version v0.1.0-draft.11, revised 24 September 2026.
- [6] Johnston, Daniel R., and Yang, Andrew, Some explicit estimates for the error term in the prime number theorem, *Journal of Mathematical Analysis and Applications* **527** no. 2 (2023) Article 127460, doi:10.1016/j.jmaa.2023.127460. Theorem 1.1; preprint version 2, 20 April 2022.
- [7] Mossinghoff, Michael J., and Trudgian, Timothy S., Nonnegative trigonometric polynomials and a zero-free region for the Riemann zeta-function, *Journal of Number Theory* **157** (2015) 329–349, doi:10.1016/j.jnt.2015.05.010. Theorem 1, preprint p. 3; version 1, 15 October 2014.
- [8] Qian, Tao, Xu, Yuesheng, Yan, Dunyan, Yan, Lixin, and Yu, Bo, Fourier spectrum characterization of Hardy spaces and applications, *Proceedings of the American Mathematical Society* **137** no. 3 (2009) 971–980, doi:10.1090/S0002-9939-08-09544-0.
- [9] National Institute of Standards and Technology, NIST Digital Library of Mathematical Functions, <https://dlmf.nist.gov/>. Equations 5.9.16, 5.11.2, 5.11.13 and 25.2.8.
- [10] Jury, Michael T., Martin, Robert T. W., and Shamovich, Eli, Blaschke–singular–outer factorization of free non-commutative functions, *Advances in Mathematics* **384** (2021) Article 107720, doi:10.1016/j.aim.2021.107720. Corollary 5.11; preprint version 2, 4 February 2020.